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ORIGINAL STUDY

Valuing Fruit Tree Lease Contracts Under Uncertainty: A Probabilistic Framework for Fair Pricing

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ABSTRACT

Fruit tree lease contracts are a prevalent economic practice in Indonesia, especially within rural communities. This study addresses the challenge of establishing equitable contract prices for both lessees and tree owners, specifically by integrating the inherent uncertainties associated with crop yield and fruit price fluctuations. To achieve this, we develop and employ two distinct models: Fixed-Time Discount Model (FTD) and the Dynamic-Time Discount Model (DTD). Each model is mathematically formulated, leveraging a Poisson distribution to capture yield uncertainty and a Uniform distribution to represent fruit price variability. Through computations, we evaluate the impact of key parameters – average yield (λ) and harvest limits (K_N) – on contract pricing and the resulting net lessee profits. Our findings reveal that the DTDM exhibits greater sensitivity to parameter variations compared to the other models. This research provides a practical and robust framework for structuring fair and efficient fruit tree lease contracts, contributing to more equitable agricultural agreements.

Keywords: Fruit tree leasing, Probabilistic modeling, Poisson distribution, Option-based valuation, Agricultural contract pricing

1. Introduction

Global uncertainty has affected many industries, making emerging market and developing economies (EMDEs) more exposed to abrupt fluctuations in commodity prices [1]. Price volatility and harvest yield uncertainty are major issues for farmers and market participants in the agriculture industry. Factors such as fluctuating weather patterns, pest infestations, and global market dynamics present substantial risks to farm incomes and the sustained stability of agricultural output. Risk management in this context increasingly draws from insights across diverse sectors—such as FinTech [2], energy markets [3], and weather forecasting [4]—where predictive modeling and data-driven strategies have been successfully applied to navigate volatility and uncertainty. A variety of financial instruments have been devised in agriculture, including futures contracts, crop insurance, and probabilistic pricing models based on projected outcomes [5, 6]. While these approaches offer solutions for certain commodities, many aspects of agricultural systems remain unaddressed, particularly contract schemes involving productive assets like fruit-bearing trees.

Fruit tree lease contracts are a common transactional type in Indonesia's rural agricultural sector,

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notably for fruit crops. In this arrangement, a lessee is granted the right to pick fruit from specific trees for a set duration. This technique provides operational freedom to lessees by eliminating the requirement for land ownership and initial cultivation, while also allowing tree owners to generate passive revenue without extensive management duties. However, one of the key challenges of this system is determining an equitable contract price for both lessors and lessees, particularly given the uncertainty surrounding harvest yields and market pricing. To present, systematic valuation models that explicitly account for the dynamic uncertainties of fruit tree lease contracts are scarce, prompting the development of more adaptive and market-responsive valuation methods.

The option valuation approach provides a robust framework for managing uncertainty in contract design. In financial theory, an option is a derivative contract that grants the holder the right—but not the obligation—to buy or sell an underlying asset at a predetermined price, known as the strike price, within a specified time frame called the expiration date [7, 8]. Based on the nature of the right, options are generally classified into two types: a call option, which gives the right to purchase an asset, and a put option, which gives the right to sell it [5].

In pricing these instruments, the option's value is typically computed as the present value of its expected payoff, discounted at the risk-free interest rate. A foundational method for valuing European-style call options was developed by [9], which assumes asset prices follow a log-normal stochastic process and that markets are free of arbitrage. Under this model, the price of a European call option is given by the expected value of its payoff at maturity, discounted to present value, is expressed in Eq. (1):

$$C = e^{-rT} \cdot \mathbb{E}[\max(S_T - K, 0)], \tag{1}$$

where C is the option price, S_T is the asset price at maturity, and K is the strike price. The closed-form solution to this expectation, known as the Black-Scholes formula, is expressed in Eqs. (2) and (3):

$$C = S_0 N(d_1) - K e^{-rT} N(d_2), \tag{2}$$

where:

$$d_1 = \frac{\ln(S_0/K) + (r + \sigma^2/2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \tag{3}$$

In this formula, S_0 represents the current price of the underlying asset, r is the risk-free interest rate, σ denotes the volatility of the asset, and $N(\cdot)$ is the cumulative distribution function of the standard normal distribution. This model forms the basis for many extensions in option-based valuation and provides a useful analogy for pricing uncertain outcomes in non-financial contexts, such as agricultural leasing or contract-based resource management.

This conceptual framework has found extensive application within the agricultural sector, extending beyond conventional commodity price hedging strategies to encompass risk management in investment decisions, land utilization optimization, and adaptation strategies in the face of climate change [10]. Real Options Analysis (ROA), for instance, has been effectively deployed by USAID in the evaluation of agricultural research and development initiatives, demonstrating a superior alternative to the less adaptable Expected Net Present Value (ENPV) methodology in contexts characterized by investment uncertainty [10]. Furthermore, in the domain of agricultural technology (AgTech) startup valuation, ROA has been shown to outperform Discounted Cash Flow (DCF) analysis due to its capacity to quantify growth prospects and managerial flexibility [6]. Moreover, option-based approaches have been leveraged to facilitate informed decision-making by farmers in response to price fluctuations and policy shifts concerning agricultural land conversion to bioenergy, thereby addressing the static limitations inherent in Net Present Value (NPV) assessments [11]. Additionally, in confronting the challenges posed by climate change, option valuation techniques are increasingly applied to evaluate the risk of stranded assets, such as agricultural land and infrastructure vulnerable to environmental perturbations [12].

Building upon this foundation, the present study endeavors to apply the option valuation framework to fruit tree lease contracts, an area demonstrably lacking in comprehensive valuation models. By adapting option pricing concepts to these unique agricultural contracts, this research aims to develop a robust methodology for determining equitable contract prices, explicitly incorporating both the inherent uncertainty of harvest yields and the pronounced volatility of fruit prices within relevant markets. This study introduces two option-based approaches: the Fixed-Time Discounting Model (FTDM) and the Dynamic-Time Discounting Model (DTDM). The FTDM employs a static discount factor to assess contract value based on expected harvest yields. The DTDM integrates a geometric distribution to capture the stochasticity inherent in contract duration. Through these methodological innovations, this study aims to address the identified lacunae in the valuation literature concerning fruit tree lease contracts and to furnish practical guidelines for the formulation of equitable and efficient contractual agreements for agribusiness stakeholders.

Consistent with option pricing principles, the determination of a fruit tree lease contract price in

this study is predicated on the expectation of future harvest yields, explicitly accounting for the dual uncertainties of yield quantity and fruit market price. Within the option valuation paradigm, crop yields are conceptualized as the underlying asset, with the anticipated profit stream serving as the foundation for contract price calculation. However, key distinctions differentiate conventional stock options from the nuanced context of fruit tree leases. While stock options typically grant the holder the right to exercise at a discrete maturity date, fruit tree lease contracts, as examined herein, present a dynamic framework wherein contracts may terminate within a single period or extend across multiple periods, contingent upon realized harvest outcomes. Specifically, should the harvest yield fall below a pre-agreed threshold, the lessee retains the option to extend the contract for a subsequent period, subject to a predetermined yield-sharing arrangement. This nuanced approach facilitates the establishment of a contract price deemed mutually equitable, optimizing net benefits for both contracting parties.

2. Contract valuation

This section introduces the basic structure and assumptions used to model a fruit tree lease contract in a probabilistic framework. The model incorporates two main stochastic variables—yield and fruit price—representing the quantity of output and its economic value. By formalizing these components through appropriate probability distributions, this section lays the groundwork for analyzing the dynamics of the lease contract, including the mechanism for contract renewal based on yield thresholds and the application of discount factors to account for the time value of money.

2.1. Probabilistic model of tree lease contracts

In this research, tree lease contracts are modeled by considering two primary stochastic variables: harvest yield (N_T) and fruit price (P_T), forming a discrete-time bivariate stochastic process $\{(N_T, P_T) : T \in \mathbb{N}\}$. The harvest yield ($N_T$) represents the quantity of harvest in period T (e.g., number of fruits or kilograms), while the fruit price (P_T) reflects the market price per unit in the same period. The joint sample space for this stochastic pair is defined in Eq. (4), where:

$$\Omega = \mathbb{N}_0 \times [a, b], \tag{4}$$

where $\mathbb{N}_0$ denotes the set of non-negative integers (for discrete harvest yields), and $[a, b]$ is a continuous and bounded interval for fruit prices.

Further, the model posits the following probabilistic distributions for these stochastic variables:

- **Harvest Yield Probability Distribution (N_T):** The harvest yield (N_T) is assumed to follow a Poisson distribution with intensity parameter λ, representing the average yield per period. The corresponding probability mass function is given in Eq. (5):

$$P_N(N_T = n) = \frac{\lambda^n e^{-\lambda}}{n!}, \quad n \in \mathbb{N}_0 \tag{5}$$

- **Fruit Price Probability Distribution (P_T):** The fruit price (P_T) is assumed to be continuously and uniformly distributed over the interval $[a, b]$. The probability density function is shown in Eq. (6):

$$f_P(P_T = p) = \begin{cases} \frac{1}{b-a}, & \text{if } p \in [a, b] \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

As an initial simplification, the model assumes that the harvest yield (N_T) and fruit price (P_T) variables are mutually independent. This assumption can be relaxed in further model developments. These distributional assumptions were selected based on preliminary field data collected in December 2024. Observations indicate that the number of harvested fruits per tree tends to follow a Poisson-like variability, consistent with biological yield processes. Likewise, fruit prices at the farmer level fluctuate within a stable and bounded range without pronounced skewness, justifying the use of a uniform distribution. Although these assumptions introduce simplifications, they provide a reasonable and analytically tractable approximation of typical trading patterns observed in rural durian markets in Indonesia.

2.2. Tree lease contracts with threshold

The contract extension mechanism in this model is contingent on a harvest yield threshold, denoted as K_N. This threshold K_N represents the minimum agreed-upon yield level within the contract. The contract extension conditions are formulated as follows:

- **Case 1: Low Harvest Yield ($N_T < K_N$)** If the harvest yield in period T is below the threshold K_N, the contract is automatically extended for one additional period. In this extension scenario, the harvest yield in the extension period is shared equally between the lessee and the tree owner. Consequently, the lessee's payoff in the extension period becomes $\frac{1}{2}N_T P_T + C$, where C represents the initial contract price or the intrinsic value of the contract. The addition of value C intrinsically

arises in the extension context due to the recursive nature of the extended contract, representing the value of the embedded extension option.

- **Case 2: High Harvest Yield ($N_T \geq K_N$)** If the harvest yield in period T is equal to or exceeds the threshold K_N, the lessee receives the entire harvest yield ($N_T P_T$), and the lease contract terminates without further extension.

It is important to note that this initial model formulation does not explicitly limit the maximum number of contract extensions. In real-world practices, tree lease contracts might incorporate limitations on the extension period.

2.3. Discount factor in the time value of money

The Time Value of Money (TVM) concept is a fundamental principle in finance asserting that money today is worth more than the same nominal amount in the future [13, 14]. This value differential arises due to the potential for investing present money to generate returns over time. To account for changes in monetary value over time, a *discount factor* is employed [15]. In the context of tree lease contract valuation, the discount factor will be integrated into the option pricing model to reflect the time value of future harvest income. The selection of an appropriate type of discount factor will be a crucial consideration in developing more realistic fruit tree lease contract models.

3. Contract price determination models

This chapter details the mathematical approaches employed to determine the contract price (C), which constitute the core methodological contribution of this research. We explore two distinct models based on discount factors: the Fixed-Time Discount (FTD) Model and the Dynamic-Time Discount (DTD) Model

3.1. Contract definition and profit function

Within the stipulations of the contract, it is acknowledged that contract extensions may occur up to a pre-defined maximum extension limit. Once the contract period reaches this maximum extension, no further extensions will be granted, even if the harvest yield $N_T < K_N$. Furthermore, the contract price is computed as the present value of the expected profit, as shown in Eq. (7):

$$C = e^{-rT} \cdot \mathbb{E}[f_T]. \tag{7}$$

This valuation principle forms the basis for modeling the fruit tree lease price. Following the previously established rules on yield-sharing between lessee and owner, the lessee's payoff f_T is defined piecewise in Eq. (8), depending on whether the observed yield meets the contractual threshold:

$$f_T = \begin{cases} \frac{1}{2} N_T P_T + C, & \text{if } N_T < K_N \\ N_T P_T, & \text{if } N_T \geq K_N, \end{cases} \tag{8}$$

where N_T is the harvest quantity at maturity, which can be the number of fruits or kilograms of fruit yield, K_N is the minimum yield threshold for contract termination or extension into the next period, P_T is the fruit price at maturity, and C is the contract price for the extended fruit tree lease.

In fruit tree lease contracts, it is self-evident that the lessee needs to undertake maintenance and security efforts to ensure optimal fruit production from the trees. All these endeavors naturally incur costs (c_0), representing the tree maintenance expenses borne by the lessee throughout the lease period. These maintenance costs encompass fertilization, irrigation, and spraying, among others. Therefore, the contract price, when considering the net profit for the lessee, is given by Eq. (9):

$$C = e^{-rT} \cdot \mathbb{E}[f_T - c_0 - C] = e^{-rT} \cdot \left(\mathbb{E}[f_T] - c_0 - C\right), \tag{9}$$

which represents the gross profit reduced by the cost and the lease premium or contract price itself. While this principle is not employed in standard options theory, it is intuitively more suitable when aiming to determine a fair contract price based on the lessee's net profit. Consequently, we can formally define the contract price in fruit tree leases as follows:

Definition 1 (Formal Definition of Contract Price)**:** The contract price for fruit tree leases is defined as the present value of the lessee's net profit. This price is a function of several factors, including probability parameters, costs, and discount factors. Mathematically, it can be expressed as:

$$C = f(p, r, c_0, K_N, \lambda, a, b),$$

where p represents the probability that the harvest yield N_T meets or exceeds the minimum threshold K_N. The parameter r denotes the risk-free interest rate, while c_0 accounts for maintenance costs incurred by the lessee. The threshold K_N sets the minimum required yield for contract termination or extension. Additionally, λ represents the expected average yield,

and the parameters a and b define the lower and upper bounds of the fruit price at maturity.

3.2. Fixed-time discount model (FTD)

The Fixed-Time Discount Model (FTDM) offers a valuation approach for fruit tree lease contracts under a fixed time horizon T, capturing yield uncertainty through conditional expectations. This model considers the crop yield N_T and fruit price P_T, with two scenarios based on whether the harvest yield falls below or meets/exceeds a minimum threshold K_N. The respective expected profits are denoted in Eq. (10):

$$\xi_1 := \mathbb{E}\left[N_T P_T \mid N_T < K_N\right], \quad \xi_2 := \mathbb{E}[N_T P_T \mid N_T \geq K_N], \tag{10}$$

where ξ_1 is the expected profit under low yield and ξ_2 under high yield. Let $p = \mathbb{P}(N_T \geq K_N)$ be the probability that the yield meets or exceeds the threshold, and $q = 1 - p$ the probability that it does not.

1. **Conditional Expectations of Yield:** The expected yield conditional on whether the threshold K_N is met or not can be expressed in Eqs. (11) and (12):

$$\mathbb{E}[N_T \mid N_T < K_N] = \sum_{n=0}^{K_N-1} \frac{n \cdot \mathbb{P}(N_T = n)}{\mathbb{P}(N_T < K_N)}, \tag{11}$$

$$\mathbb{E}[N_T \mid N_T \geq K_N] = \sum_{n=K_N}^{\infty} \frac{n \cdot \mathbb{P}(N_T = n)}{\mathbb{P}(N_T \geq K_N)}. \tag{12}$$

   These expectations rely on the assumption that N_T follows a Poisson distribution.
2. **Expected Profits:** Assuming the fruit price P_T is uniformly distributed over the interval $[a, b]$ and is independent of N_T, the expected price follows Eq. (13):

$$\mathbb{E}[P_T] = \frac{a+b}{2}. \tag{13}$$

   Hence, the expected profits under each condition follows the Eqs. (14) and (15):

$$\xi_1 = \frac{a+b}{2} \cdot \sum_{n=0}^{K_N-1} \frac{n \cdot \mathbb{P}(N_T = n)}{\mathbb{P}(N_T < K_N)}, \tag{14}$$

$$\xi_2 = \frac{a+b}{2} \cdot \sum_{n=K_N}^{\infty} \frac{n \cdot \mathbb{P}(N_T = n)}{\mathbb{P}(N_T \geq K_N)}. \tag{15}$$

3. **Expected Profit from the Contract:** Applying the Law of Total Expectation, the overall expected profit is computed by weighting each conditional expectation by its respective probability, as in Eq. (16):

$$\mathbb{E}[f_T] = q \cdot \frac{1}{2}\xi_1 + p \cdot \xi_2. \tag{16}$$

4. **Contract Price Valuation:** As introduced earlier in Eq. (9), the contract price C represents the discounted value of the lessee's net expected profit, which incorporates future payoffs, maintenance costs, and the contract premium itself. By substituting the total expected profit $\mathbb{E}[f_T]$ from Eq. (16) into that expression and rearranging terms, the contract price can be derived in closed form as shown in Eq. (17):

$$\begin{aligned} C &= e^{-rT}\left(\frac{1}{2}(1-p)\xi_1 + C(1-p) + p\xi_2 - c_0 - C\right) \\ &= \frac{e^{-rT}\left(\frac{1}{2}(1-p)\xi_1 + p\xi_2 - c_0\right)}{1 + pe^{-rT}}. \end{aligned} \tag{17}$$

Eq. (17) provides the final contract price incorporating expected yields, prices, discounting, and maintenance cost c_0 under a fixed time horizon. This model serves as a benchmark before transitioning to dynamic contract durations in subsequent sections. The final formula for the fair contract price C under the Fixed-Time Discount Model captures several important elements: it accounts for probabilistic expectations from both low- and high-yield scenarios, incorporates the time value of money through discounting, assumes independence between yield and price, and adjusts for contract risk and penalties. This formulation operates under the assumption of a fixed contract period T, typically set as one year, and applies a continuous discount factor to calculate the present value of expected profits. However, this simplification may not fully capture the dynamic nature of time in financial modeling, particularly in contexts involving inflation, liquidity constraints, and risk exposure [16]. When the actual duration of the contract is uncertain or potentially indefinite, the fixed-time approach may lead to inaccurate pricing. To address this limitation, the next section introduces a stochastic model in which the contract duration T follows a Geometric distribution, allowing for a more realistic and flexible valuation framework.

3.3. Dynamic-time discount model (DTD)

The Fixed-Time Discount (FTD) model, while foundational, assumes a constant contract duration for

discounting, neglecting the inherent stochasticity of contract length in tree lease agreements. The Dynamic-Time Discount (DTD) model addresses this limitation by explicitly modeling the contract length as a random variable, making it more reflective of real-world scenarios where the duration of the lease is uncertain. The core idea is to incorporate the Time Value of Money more accurately by allowing the discount period to be dynamic and dependent on when the contract terminates.

The DTD model centers around the concept of contract length (T) as a random variable. We define T as the number of periods until the accumulated harvest yield (N_t) first meets or exceeds a predetermined threshold (K_N), triggering contract termination:

Definition 2 (Contract Length as a Random Variable)**:** The contract length T is defined as the minimum number of periods ($t \geq 1$) required for the cumulative harvest yield N_t to reach or surpass the threshold K_N:

$$T = \min\{t \geq 1 : N_t \geq K_N\}.$$

A crucial aspect of the DTD model is understanding the probability distribution of this random contract length T. Under the assumption that harvest yields in each period are independent and identically distributed Poisson random variables, we can demonstrate that T follows a Geometric distribution. This is formally stated in Theorem 1.

Theorem 1 (Distribution of Contract Length T)**:** *Assuming independent Poisson distributed harvest yields* $N_t \sim \text{Poisson}(\lambda)$ *and a success probability* $p = \mathbb{P}(N_t \geq K_N)$*, the contract length* T *follows a Geometric distribution with parameter* p*, denoted as* $T \sim \text{Geom}(p)$*. The success probability* p *is calculated as* $p = 1 - \sum_{n=0}^{K_N-1} \frac{\lambda^n e^{-\lambda}}{n!}$.

The probability of the contract lasting exactly k periods is given by the Geometric probability mass function (PMF):

$$\mathbb{P}(T = k) = (1-p)^{k-1} \cdot p, \quad k = 1, 2, 3, \ldots. \tag{18}$$

The expected or average contract length, a key characteristic, is then simply the expectation of a Geometric distribution:

$$\mathbb{E}[T] = \frac{1}{p}. \tag{19}$$

Lessee's profit structure under the dynamic-time discount model (DTD)

This section outlines the structure of the lessee's profit under the DTDM framework, where the contract may extend over multiple periods until a minimum harvest threshold K_N is reached. The total discounted net profit $\tilde{f}_T$ is composed of sequential stages, which are described as follows:

1. **Profit in the Initial $T-1$ Periods:**
   During the first $T-1$ periods, where the harvest yield N_T is consistently below K_N, the lessee receives only half of the harvest value. This partial income is denoted as ζ_1, and its expected value is defined in Eq. (20):

$$\mathbb{E}[\zeta_1] = \frac{\xi_1}{2}. \tag{20}$$

2. **Profit in the Final Period T:**
   When the threshold is finally met or exceeded in period T, the lessee obtains the full harvest, denoted by ζ_2. Its expected value is given in Eq. (21):

$$\mathbb{E}[\zeta_2] = \xi_2. \tag{21}$$

3. **Total Net Profit over the Lease Duration:**
   Summing the discounted values of ζ_1 and ζ_2 over their respective periods, subtracting the maintenance cost c_0 per period and the contract price C, we arrive at the total net profit $\tilde{f}_T$ as shown in Eq. (22):

$$\tilde{f}_T = \underbrace{\sum_{i=1}^{T-1} (\zeta_1 - c_0) \cdot e^{-ri}}_{\text{Profit during periods 1 to } T-1} + \underbrace{(\zeta_2 - c_0) \cdot e^{-rT}}_{\text{Profit in period } T} - C \cdot e^{-rT}. \tag{22}$$

4. **Expected Contract Price:**
   The expected value of C can be computed by taking the expectation over $\tilde{f}_T$ in Eq. (22). The resulting expression is given in Eq. (23):

$$C = \mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt} (\mathbb{E}[\zeta_1] - c_0)\right] + \mathbb{E}\left[e^{-rT} (\mathbb{E}[\zeta_2] - c_0 - C)\right]. \tag{23}$$

5. **Simplified Contract Price using Discount Expectations:**
   Assuming independence between T, N_T, and P_T, Eq. (23) simplifies to the closed-form expression

shown in Eq. (24):

$$
\begin{aligned}
C &= \mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right](\mathbb{E}[\zeta_1]-c_0)+\mathbb{E}\left[e^{-rT}\right](\mathbb{E}[\zeta_2]-c_0-C) \\
&= \frac{\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right](\mathbb{E}[\zeta_1]-c_0)+\mathbb{E}\left[e^{-rT}\right](\mathbb{E}[\zeta_2]-c_0)}{1+\mathbb{E}\left[e^{-rT}\right]} \\
&= \mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right]\left(\frac{\xi_1}{2}-c_0\right)+\mathbb{E}[e^{-rT}](\xi_2-c_0-C) \\
&= \frac{\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right]\left(\frac{\xi_1}{2}-c_0\right)+\mathbb{E}[e^{-rT}](\xi_2-c_0)}{1+\mathbb{E}[e^{-rT}]}.
\end{aligned}
\tag{24}
$$

This formulation mirrors the net-present-value concept introduced earlier in Eq. (9), and reflects the economic rationality of balancing risk, time, and reward in lease pricing.

Computing Discount Expectations and Final Pricing Formula

The expectations $\mathbb{E}[e^{-rT}]$ and $\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right]$ used in Eq. (24) can be explicitly calculated under the assumption that the contract duration $T \sim \text{Geom}(p)$, with $q = 1 - p$. The following theorem provides the closed-form expression for the expected discount factor:

Theorem 2: *Let T be a random variable following a Geometric distribution with parameter $p \in (0, 1)$ and $q = 1 - p$. For a constant $r > 0$, the expectation of the continuous discount factor e^{-rT} is given by:*

$$
\mathbb{E}[e^{-rT}] = \frac{pe^{-r}}{1 - qe^{-r}},
$$

where e^{-r} is the base of the exponential discounting.

Proof: Let $g_T(t) = e^{-rt}$, then by the law of the unconscious statistician, we have:

$$
\begin{aligned}
\mathbb{E}[e^{-rT}] &= \sum_{T=1}^{\infty} g_T(t)\mathbb{P}(T = t) \\
&= e^{-r}p + e^{-2r}(1-p)p + e^{-3r}(1-p)^2 p + \cdots \\
&\quad \text{(Geometric series with ratio } e^{-r}(1-p) < 1) \\
&= \frac{pe^{-r}}{1-(1-p)e^{-r}} = \frac{pe^{-r}}{1-qe^{-r}}.
\end{aligned}
$$

1. **Deriving the Expected Discount Factor $\mathbb{E}[e^{-rT}]$**
   From Theorem 2, we directly obtain the closed-form expression:

$$
\mathbb{E}[e^{-rT}] = \frac{pe^{-r}}{1 - qe^{-r}}. \tag{25}
$$

2. **Deriving the Expected Sum of Discounted Values $\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right]$:**
   Consider the finite geometric summation as in the following Eq. (26):

$$
\sum_{t=1}^{T-1} e^{-rt} = e^{-r} + e^{-2r} + \cdots + e^{-r(T-1)}, \tag{26}
$$

which simplifies to:

$$
\sum_{t=1}^{T-1} e^{-rt} = \frac{e^{-r} - e^{-rT}}{1 - e^{-r}}. \tag{27}
$$

Taking expectations on both sides of Eq. (27), we derive:

$$
\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right] = \frac{e^{-r}}{1-e^{-r}} - \frac{1}{1-e^{-r}} \cdot \mathbb{E}[e^{-rT}]
$$

Substituting from Theorem 2, we obtain:

$$
\begin{aligned}
\mathbb{E}\left[\sum_{t=1}^{T-1} e^{-rt}\right] &= \frac{1}{1-e^{-r}}\left(e^{-r} - \frac{pe^{-r}}{1-qe^{-r}}\right) \\
&= \frac{(1-qe^{-r})e^{-r} - pe^{-r}}{(1-e^{-r})(1-qe^{-r})} \\
&= \frac{(1-qe^{-r}-p)e^{-r}}{(1-e^{-r})(1-qe^{-r})} \\
&= \frac{(q-qe^{-r})e^{-r}}{(1-e^{-r})(1-qe^{-r})} \\
&= \frac{q(1-e^{-r})e^{-r}}{(1-e^{-r})(1-qe^{-r})} \\
&= \frac{qe^{-r}}{1-qe^{-r}}, \quad \text{with } r \neq 0 \text{ and } q \neq e^{r}.
\end{aligned}
$$

where the final expression follows by substituting Eq. (25).

3. **Final Closed-Form Pricing Formula**:
   Now that we have both expectations, substitute them into the simplified pricing expression

in Eq. (24):

$$C = \frac{\left(\frac{qe^{-r}}{1-qe^{-r}}\right)\left(\frac{\xi_1}{2} - c_0\right) + \left(\frac{pe^{-r}}{1-qe^{-r}}\right)(\xi_2 - c_0)}{1 + \frac{pe^{-r}}{1-qe^{-r}}},$$

$$r \neq 0\,, q \neq e^r,$$

This final form is captured in Eq. (28) and reflects the present value of expected profits under the Dynamic-Time Discount Model.

$$C = \frac{qe^{-r}\left(\frac{\xi_1}{2} - c_0\right) + pe^{-r}(\xi_2 - c_0)}{1 - qe^{-r} + pe^{-r}}, \quad r \neq 0\,, q \neq e^r. \tag{28}$$

This approach better incorporates the concept of the Time Value of Money compared to the previous model. This formula reflects the present value of expected profits under dynamic contract durations and connects back to the net present value framework outlined in Eq. (9). The Dynamic-Time Discount Model (DTD) adjusts the contract price based on several interrelated factors: the probability of reaching the harvest threshold within fewer periods (higher p implies shorter contract durations), the time value of money which states that future payments are worth less in present terms, and the lessee's exposure to profit reductions during early extension periods. When the average yield λ is significantly lower than the contract threshold K_N, the likelihood p of achieving the target in any given period becomes small. This results in longer expected contract durations and amplifies the effect of discounting, thereby increasing the fair contract price C to reflect greater risk and delayed returns. Although this mechanism provides a robust analytical foundation, excessively long contracts may be impractical. In real-world applications, such agreements are typically capped or renegotiated. Nonetheless, the DTD model offers a rigorous benchmark for determining equitable pricing under uncertainty and temporal variability.

4. Results and discussion

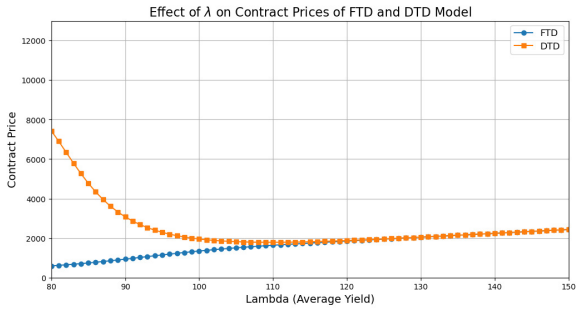
Assessing the practical relevance of the proposed models requires empirical grounding through contract pricing simulations under varying harvest yield scenarios. These simulations are designed not only to evaluate model behavior but also to inform stakeholders—such as farmers, cooperatives, and policy-makers—regarding potential financial risks or imbalances inherent in standard fixed-price leasing practices. The mathematical framework developed in the preceding sections is evaluated using parameter values derived from preliminary fieldwork conducted in December 2024, which included direct observations and interviews with Indonesian durian farmers.

Field data indicate that durian trees typically yield around 120 fruits per harvest, with individual fruit prices ranging between IDR 30,000 and IDR 50,000 at the wholesale level. Tenants are responsible for both maintenance and security, incurring average costs of IDR 1,000,000 per tree per harvest. These empirical insights serve as the foundation for parameter selection in the simulation: the harvest yield N_T is modeled using a Poisson distribution with average yield $\lambda = 120$, while the fruit price P_T is assumed uniformly distributed between $a = 30{,}000$ and $b = 50{,}000$. The contract threshold is set at $K_N = 100$, with an interest rate $r = 5\%$ and maintenance cost $c_0 = 1{,}000{,}000$ IDR. Computational simulations are conducted using Python, employing libraries such as `NumPy` and `SciPy` for probabilistic modeling and `Matplotlib` for visualization. Numerical methods are used to explore how variations in average yield and threshold affect both the calculated lease prices and the resulting profits for tenants. These simulations demonstrate the sensitivity of the contract price to yield volatility and highlight the policy implications of different contract designs under real-world agricultural conditions.

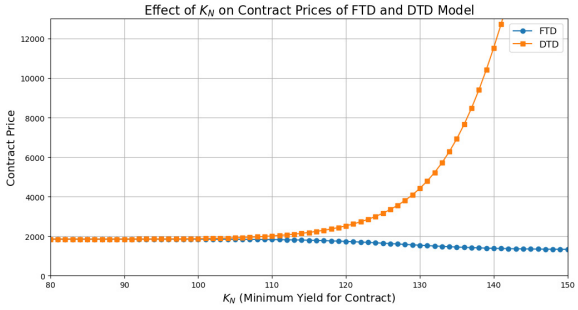
Table 1. Comparison of contract prices across FTD and DTD models.

Lambda (λ)	FTD Price	DTD Price
80	594.59	7430.92
90	935.47	3076.96
100	1346.26	1954.15
110	1638.50	1788.00
120	1851.70	1877.55
130	2242.54	2242.71
140	2632.51	2632.51

The contract prices calculated using the Fixed-Time Discount (FTD) and Dynamic-Time Discount (DTD) models are compared in Table 1. The results clearly show that the FTD model consistently yields lower contract prices across all average harvest yields (λ) presented. In contrast, the DTD model produces significantly higher prices, particularly when the average harvest yield (λ) is low. This considerable difference is attributed to the DTD model's heightened sensitivity to inherent uncertainty and the associated probability of contract extension. While the FTD model exhibits more stable prices, they remain substantially lower than those from the DTD model, reflecting a different balance between contract flexibility and discount risk. In summary, the FTD model proves suitable for low-risk scenarios, whereas the DTD model is more appropriate for conditions characterized by high uncertainty. It's also



(a) Simulation of Contract Price Changes based on λ



(b) Simulation of Contract Price Changes based on K_N

Fig. 1. Effect of (a) Average harvest yield (λ) and (b) Contract threshold (K_N) on contract prices.

important to note that for harvest yields significantly above the contract threshold, such as $\lambda = 120, 130,$ or 140, both the FTD and DTD models yield relatively similar prices. Comparing these computational results with real-world conditions, where the typical tree lease price is around IDR 1,000,000 per tree per harvest, reveals potential discrepancies that could disadvantage either farmers or tenants. For instance, under the FTD model:

- If the harvest yield falls 10 units below the contract threshold, tenants incur a loss of approximately 6.5%.
- If the harvest yield is 20 units below the contract threshold, tenant losses increase to around 40.6%.
- Conversely, if the harvest yield significantly exceeds the contract threshold, the tree owners experience gradual losses, for example, 34.6% and 63.6% for increasing over-threshold yields.

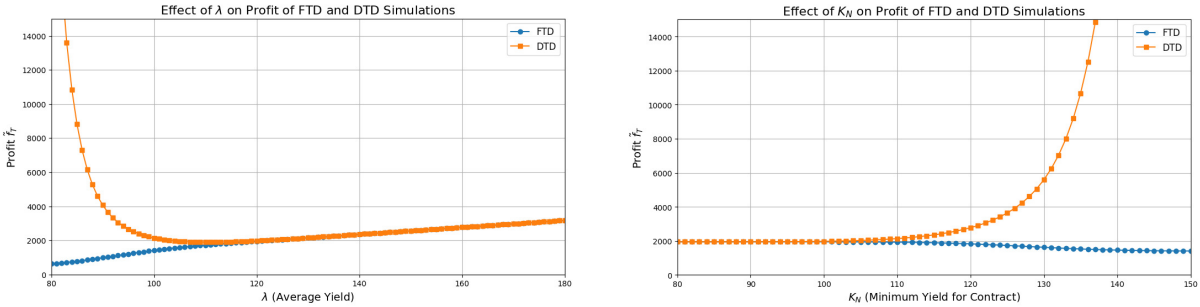
These discrepancies highlight the need for contractual mechanisms to mitigate risk. In practice, if the harvest is significantly below the threshold, the contract is often extended for the next season without additional cost to the tenant. Conversely, if the harvest substantially exceeds the threshold, tenants typically share a portion of the excess yield with the tree owner. Under the DTD model, the primary concern for tree owners arises when the computed contract price, as shown in Table 1, exceeds the typical market lease price of IDR 1,000,000. To mitigate significant losses for both tenants and tree owners, field agreements (as per contract 2.2) often stipulate that contract extensions are contingent on the harvest yield threshold:

- If the harvest yield in period T falls below the threshold K_N, the contract is automatically extended for one additional period.
- If the harvest yield in period T is equal to or exceeds the threshold K_N, the tenant receives the entire harvest ($N_T P_T$), and the lease contract concludes without further extension.

4.1. Impact of average harvest yield (λ) and contract threshold (K_N) on lease price

Understanding the factors that influence tree lease contract prices is crucial for designing optimal leasing agreements. Two key relationships are examined: the impact of average harvest yield (λ) and the effect of the contract threshold (K_N) on lease prices. The simulations were conducted using the baseline parameters, varying only λ and K_N, with results displayed in Fig. 1.

Fig. 1(a) compares the contract prices (C) derived from the FTD and DTD models (for $t = 1$ and $t = 2$) as a function of λ. Generally, contract prices in all models increase with higher average harvest yields (λ), indicating a positive correlation between expected yield and contract value. However, the DTD model shows a distinct pattern where contract prices are exceptionally high at low λ values, then decline sharply before stabilizing at higher λ. This behavior reflects the Geometric distribution's influence in the DTD model–lower λ increases the likelihood of multiple contract extensions, raising total expected profit and, consequently, contract price. Fig. 1(b) illustrates the impact of the contract threshold (K_N) on lease price. All models yield similar contract prices when K_N is close to λ, but diverge as K_N increases. The FTD model shows relatively stable and slightly decreasing prices, which is expected since a higher K_N makes contract termination more difficult. However, the DTD model exhibits a steep price increase for K_N values exceeding λ. This occurs because in the DTD model, a higher K_N reduces the probability of meeting the harvest threshold, increasing the likelihood of contract extensions



(a) Comparison of Lessee Profits against λ (b) Comparison of Lessee Profits against K_N

Fig. 2. Effect of (a) Average harvest yield (λ) and (b) Contract threshold (K_N) on lessee profits.

and subsequently raising lease prices. The sensitivity of the DTD model to K_N suggests that setting K_N values close to historical averages and limiting contract duration to one or two periods are practical strategies for maintaining affordable lease pricing.

4.2. Lessee profit

A crucial aspect of contract evaluation is the lessee's net profit ($\tilde{f}_T$) under different models. This section compares the net profits from the FTD and DTD models using both theoretical derivations.

Theorem 3 (Lessee Net Profit Formulas)**:** *Assuming the net profits for the FTD and DTD models are* $\tilde{f}_T^{(1)}$ *and* $\tilde{f}_T^{(2)}$ *respectively, and the contract prices are* $C^{(1)}$ *and* $C^{(2)}$*, respectively, the lessee's net profit can be formulated as follows:*

1. ***FTD Model:***

$$\tilde{f}_T^{(1)} = \frac{q\xi_1}{2} + p\xi_2 - c_0 - pC^{(1)}$$

2. ***DTD Model:***

$$\tilde{f}_T^{(2)} = \frac{q}{p}\left(\frac{\xi_1}{2} - c_0\right) + \xi_2 - c_0 - C^{(2)}$$

Theorem 3 is derived from the valuation equations of each model by removing the discount factor, effectively representing the undiscounted profit expectation. Using Python programs, the comparison of lessee profits across the two models relative to λ and K_N is visualized in Fig. 2.

Fig. 2(a) indicates that all models yield relatively similar profits for $\lambda > 140$. However, significant differences emerge at $\lambda < 140$. The DTD model (orange line) shows a substantial increase in profit as λ decreases. This is because lower λ values increase the probability of $N_T < K_N$ events, leading to prolonged contract extensions. The DTD model's unbounded extension feature logically results in higher profits with lower λ due to extended contract durations. Fig. 2(b) reveals that all models show relatively consistent profit values when $K_N < 100$. However, for $K_N > 100$, projected lessee profits diverge. The DTD model profit increases significantly as K_N rises because the occurrence of $N_T < K_N$ becomes more frequent, extending the contract. Overall, the behavior mirrors the effect of λ on lessee profits as depicted in Fig. 2(a). This implies that tree owners can manage both profits and contract prices by adjusting the harvest threshold K_N.

These computation insights underscore the potential of mathematical modeling to support fairer lease negotiations and contract design in agriculture. By quantifying the risks faced by both tenants and tree owners, the proposed framework can inform practical decision-making, foster more transparent revenue-sharing agreements, and provide a policy-relevant benchmark for structuring risk-mitigated agricultural leases. Embedding such mechanisms into standardized contracts would enhance fairness, reduce disputes, and strengthen trust among all stakeholders involved.

5. Conclusion

This study developed two mathematical models to determine the lease contract pricing for fruit-bearing trees, namely the Fixed-Time Discount Model (FTD) and the Dynamic-Time Discount Model (DTD). Both FTD and DTD represent contracts with indefinite extensions, allowing the lessee to continue renewing the contract as long as the extension conditions are met. The FTD model provides a straightforward solution under the assumption of a fixed contract duration; however, it disregards the stochastic nature of contract extensions. In contrast, the DTD model, which employs a geometric distribution to capture contract length, exhibits high sensitivity to variations

in the average harvest yield (λ), making it suitable for scenarios with significant uncertainty. Overall, simulations indicate that the average harvest yield (λ) and the contract threshold (K_N) significantly influence contract pricing and the lessee's net profit.

While these models contribute a theoretical framework for more equitable lease agreements, they are subject to several limitations. First, the current models assume independence between harvest yield and fruit price. This simplification may not hold in practice, where inverse relationships are often observed in agricultural markets. Future work should explore joint or copula-based distributions to better capture dependency structures. Second, although the models allow for indefinite contract extensions, this does not reflect standard contractual norms. A capped extension scheme should be incorporated in future model refinements. Third, the current framework does not include external risk factors such as weather variability, pest outbreaks, or market shocks. Introducing stochastic risk variables or scenario-based analysis would improve the model's realism.

Additionally, future research should move beyond Poisson and Uniform distribution assumptions by employing empirical distribution fitting or Bayesian updates using real-world data. Broader field validation, including various regions and crop types, is also necessary. Finally, the development of practical tools or decision-support systems based on these models could support implementation by farmers, cooperatives, and policymakers.

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Conflict of interest

The authors declare no conflict of interest.

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